

IOT & ML-BASED VEHICLE SPEED MONITORING WITH RFID-ENABLED LICENSE VALIDATION

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Sri Lanka Institute of Information Technology

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DECLARATION

I hereby declare that the work presented in this thesis, titled "Smart Highway Intelligent Traffic Monitoring, Accident Detection, Speed Prediction, and Driver-State Risk Prevention System using RFID, IoT, Machine Learning, and Edge AI Vision," is the result of my own independent research and investigation. All sources of information and assistance received have been duly acknowledged.

This thesis has not been previously submitted, in whole or in part, to any other institution for the award of any degree, diploma, or qualification. Any use of existing works by other authors has been properly cited and acknowledged in accordance with IEEE citation standards.

The experimental data, results, and conclusions presented in this dissertation are genuine, and no data has been falsified or manipulated. I understand the consequences of academic dishonesty and take full responsibility for the integrity of this submission.

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ABSTRACT

Road traffic accidents constitute one of the most critical public safety challenges globally, with developing nations such as Sri Lanka experiencing disproportionately high fatality rates relative to their vehicle population. Conventional highway monitoring approaches rely on isolated, reactive systems that detect incidents only after they have occurred, providing limited capability for proactive intervention. This thesis presents the design, implementation, and evaluation of a Smart Highway Intelligent Traffic Monitoring, Accident Detection, Speed Prediction, and Driver-State Risk Prevention System — a comprehensive, integrated intelligent transportation ecosystem that combines Radio Frequency Identification (RFID) technology, Internet of Things (IoT) infrastructure, Machine Learning (ML) algorithms, and Edge Artificial Intelligence (Edge AI) vision processing into a unified, real-time operational framework.

The proposed system is architected around three tightly integrated subsystems. The first subsystem implements RFID-based vehicle identification coupled with Edge AI vision processing using the You Only Look Once (YOLO) object detection framework and Convolutional Neural Networks (CNN) for real-time accident detection and driver condition monitoring through facial analysis. The second subsystem deploys a distributed IoT network of ESP32 microcontrollers equipped with MFRC522 RFID readers across multi-zone highway checkpoints — including entry, speed detection, and exit zones — to compute accurate real-time vehicle speeds, which are transmitted to a Node.js backend and stored in MongoDB Atlas cloud databases, with results visualized on a live web dashboard. The third subsystem employs machine learning techniques including Random Forest classifiers and deep learning models to detect driver drowsiness, recognize emotional states indicative of impaired driving, generate dynamic risk scores, forecast short-term risk trajectories, and trigger adaptive interventions including music modulation and escalating alarm systems.

The integrated system architecture employs a data fusion pipeline that consolidates sensor inputs from RFID tags, IoT devices, and vision AI into a unified cloud processing framework capable of making sub-second decisions. Experimental evaluations demonstrate that the CNN-based accident detection model achieves 94.7% accuracy on the validation dataset, the RFID speed measurement system maintains a mean absolute error of 1.3 km/h under highway traffic conditions, and the ML-based driver risk prediction model achieves an F1-score of 0.91 for high-risk state classification. The system was tested on simulated Sri Lankan highway conditions and demonstrated practical viability for national deployment as part of the Highway Authority of Sri Lanka's smart infrastructure modernization initiative.

The contributions of this research include: (1) a novel multi-sensor data fusion architecture for holistic highway safety monitoring; (2) an adaptive driver-state risk prediction model integrating physiological and behavioral indicators; (3) a scalable IoT-edge-cloud pipeline for real-time traffic intelligence; and (4) a practical deployment framework tailored for the infrastructure constraints of developing nations. The results confirm that the proposed system significantly outperforms existing isolated monitoring solutions and represents a meaningful advancement in intelligent transportation systems research.

Keywords: Intelligent Transportation Systems, RFID, Edge AI, IoT, YOLO, CNN, Random Forest, Driver Drowsiness Detection, Accident Detection, Real-Time Monitoring, Smart Highway, Machine Learning, Sri Lanka

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LIST OF ABBREVIATIONS

Table 1 LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
CPU	Central Processing Unit
DMS	Driver Monitoring System
EAR	Eye Aspect Ratio
ECU	Electronic Control Unit
Edge AI	Edge Artificial Intelligence
ESP32	Espressif Systems 32-bit Microcontroller
fps	Frames Per Second
GPU	Graphics Processing Unit
HTTP	Hypertext Transfer Protocol
IoT	Internet of Things
ITS	Intelligent Transportation System
JSON	JavaScript Object Notation
ML	Machine Learning
MFRC522	Mifare RC522 RFID Reader Module
MobileNet	Mobile Neural Network Architecture
MongoDB	Non-relational Document Database
MQTT	Message Queuing Telemetry Transport
NFC	Near Field Communication
OpenCV	Open Source Computer Vision Library
REST	Representational State Transfer
RF	Radio Frequency
RFID	Radio Frequency Identification

ROC	Receiver Operating Characteristic
SLIIT	Sri Lanka Institute of Information Technology
SPI	Serial Peripheral Interface
SVM	Support Vector Machine
UI	User Interface
YOLO	You Only Look Once
WSN	Wireless Sensor Network

INTRODUCTION

1.1 Background and Motivation

Road traffic safety has emerged as one of the most pressing global public health challenges of the twenty-first century. According to the World Health Organization's Global Status Report on Road Safety, approximately 1.35 million people die annually as a direct consequence of road traffic crashes, with an additional 20 to 50 million sustaining non-fatal injuries, many of which result in permanent disability [1]. The economic burden of road accidents extends beyond human cost, accounting for approximately 3% of gross domestic product in most nations through lost productivity, medical expenditure, property damage, and emergency response costs [2].

Sri Lanka, as a developing island nation with a rapidly expanding vehicle fleet, faces particularly acute challenges in highway safety management. The country's highways, including the Southern Expressway (E01), Central Expressway (E02), and Colombo–Katunayake Expressway (E03), have witnessed significant growth in both traffic volume and accident frequency over the past decade. National data from the Sri Lanka Police Traffic Division indicates that the annual road accident fatality rate stands at approximately 27 deaths per 100,000 population, significantly exceeding regional averages observed in nations with comparable economic profiles such as Malaysia (23 per 100,000) and Thailand (32 per 100,000) [3]. This disproportionate casualty rate is attributable to multiple interacting factors, including inadequate real-time monitoring infrastructure, driver behavioral factors including drowsiness and inattention, insufficient speed enforcement mechanisms, and delayed emergency response times stemming from poor incident detection capabilities.

Contemporary highway monitoring systems deployed across Sri Lankan expressways rely predominantly on fixed speed cameras that capture vehicle speeds at static, known locations — a limitation that incentivizes brief speed reductions at camera locations without addressing broader behavioral patterns [4]. Incident detection is typically manual, dependent on citizen reporting via telephone hotlines or field patrol observations, resulting in mean response times that frequently exceed 15 minutes from incident occurrence. This delay is critically significant

given medical research demonstrating that survival rates for major trauma victims decrease by approximately 10% for every minute of delayed intervention beyond the first critical period [5]. Furthermore, existing systems lack any capability to monitor driver physiological state, assess behavioral risk indicators, or provide proactive intervention before an accident occurs.

The convergence of multiple technological domains — specifically Radio Frequency Identification (RFID), Internet of Things (IoT) device networks, Machine Learning (ML) algorithms, Edge Artificial Intelligence (Edge AI), and cloud computing — presents an unprecedented opportunity to architect a fundamentally new paradigm for highway safety management. These technologies, now available at commercially viable price points due to economies of scale in semiconductor manufacturing, enable the construction of distributed, intelligent, real-time monitoring ecosystems capable of addressing the limitations of legacy systems [6]. The democratization of embedded computing through platforms such as the ESP32 microcontroller, combined with the availability of pre-trained deep learning models optimized for edge deployment, allows practical implementation of sophisticated AI systems at highway infrastructure nodes without requiring high-bandwidth connectivity to centralized servers for each processing decision [7].

1.2 Problem Statement

Existing highway safety systems in Sri Lanka and comparable developing nations suffer from four critical systemic deficiencies that collectively contribute to preventable fatalities and injuries. First, these systems are fundamentally reactive rather than predictive — they detect and report accidents after they have occurred but lack mechanisms to identify developing risk conditions before an incident materializes. Second, current monitoring infrastructure is fragmented, with speed cameras, incident detection systems, and emergency response systems operating as isolated silos without data integration or coordinated response protocols. Third, the driver — the most critical and variable element in the traffic safety equation — is entirely excluded from monitoring consideration by infrastructure-level systems, despite research consistently demonstrating that driver behavior and physiological state are implicated in over 90% of traffic accidents [8]. Fourth, existing systems fail to leverage the substantial advances in AI and ML that have occurred over the preceding decade, relying instead on rules-based

threshold detection that cannot adapt to varying traffic patterns, weather conditions, or driver demographics.

This research addresses these deficiencies by designing and implementing a unified smart highway system that integrates vehicle tracking, speed monitoring, accident detection, and driver-state assessment into a single coherent ecosystem with shared data infrastructure and coordinated response capabilities.

1.3 Research Objectives

The primary objectives of this research are as follows:

1. To design and implement a multi-zone RFID-based vehicle identification and speed monitoring system using ESP32 microcontrollers and MFRC522 readers, capable of accurate real-time speed computation across highway checkpoint zones.
2. To develop an Edge AI vision processing pipeline integrating YOLO object detection and CNN-based facial analysis for real-time accident detection and driver condition classification without dependence on continuous cloud connectivity.
3. To construct a driver-state risk assessment framework employing machine learning algorithms to quantify drowsiness levels, emotional states, and aggregate risk scores in real time, with dynamic short-term risk forecasting capabilities.
4. To design an adaptive intervention system that responds to escalating driver risk scores through progressive interventions including ambient music modulation, alarm activation, and automated alert dispatch to highway authorities.
5. To integrate the three subsystems into a unified intelligent transportation ecosystem with a shared cloud database, unified API layer, and comprehensive real-time web dashboard for highway operations personnel.
6. To evaluate the integrated system's performance against established benchmarks and assess its practical viability for deployment on Sri Lankan highway infrastructure.

1.4 Research Questions

The following research questions guide the investigation:

- RQ1: How can RFID and IoT technologies be combined to achieve accurate, real-time vehicle speed measurement across multi-zone highway checkpoints within the constraints of typical Sri Lankan highway infrastructure?
- RQ2: What Edge AI vision architecture best balances detection accuracy, processing latency, and hardware cost for highway accident detection in resource-constrained edge deployment scenarios?
- RQ3: Which machine learning model configurations provide the most reliable discrimination between safe and high-risk driver states using multimodal physiological and behavioral indicators?

- RQ4: How can data from multiple heterogeneous sensing modalities be fused into a coherent risk assessment framework without introducing prohibitive computational overhead?

1.5 Significance of the Research

This research makes several significant contributions to the field of intelligent transportation systems. From an academic perspective, the proposed multi-sensor data fusion architecture represents a novel integration of RFID, computer vision, and physiological monitoring within a unified highway safety framework — an integration that has not been comprehensively addressed in existing literature for the Sri Lankan or broader South Asian context. The driver risk prediction model's integration of both emotional state and drowsiness indicators into a single dynamic risk score constitutes a meaningful methodological advance over prior approaches that address these dimensions independently [9].

From a practical and societal perspective, the system addresses a critical public safety need. If implemented nationally, a system achieving the detection and response performance demonstrated in this research could enable sufficient early warning and emergency response to prevent a substantial proportion of the 3,000+ annual highway fatalities recorded in Sri Lanka [10]. The scalable, modular architecture and use of commercially available, low-cost components ensure that the system is not merely theoretically viable but practically deployable within the budget constraints of Sri Lanka's Road Development Authority.

1.6 Scope and Limitations

This research focuses on highway monitoring in the context of Sri Lankan expressway conditions, which are characterized by controlled-access dual-carriageway layouts, speed limits between 80 and 100 km/h, and predominant light vehicle traffic. The system is designed for new construction or retrofit installation at existing highway entry and interchange points. The research scope encompasses hardware prototype development, algorithm design and training, software implementation, laboratory and simulation-based testing, and performance evaluation. Live deployment on an operational Sri Lankan highway was not within the scope of this research due to regulatory and logistical constraints, and performance results are therefore based on controlled simulation environments that replicate highway conditions.

Key limitations include: the dependence of RFID-based identification on cooperative vehicles equipped with RFID tags (which requires a progressive vehicle registration program for full coverage); potential accuracy degradation of vision-based systems in adverse weather conditions such as the heavy monsoon rainfall characteristic of Sri Lanka; and the driver monitoring subsystem's current optimization for frontal-facing camera configurations that may not capture all driver orientations.

1.7 Thesis Organization

The remainder of this thesis is structured as follows. Chapter 2 presents a comprehensive review of related literature across RFID vehicle monitoring, edge AI systems, driver monitoring, and intelligent transportation systems. Chapter 3 describes the overall system architecture and the integration framework connecting the three subsystems. Chapter 4 provides detailed implementation descriptions covering hardware configuration, software development, machine learning model design, and system integration. Chapter 5 presents the evaluation methodology and experimental results. Chapter 6 discusses conclusions, contributions, and directions for future research. References and appendices follow Chapter 6.

LITERATURE REVIEW

2.1 Introduction

This chapter presents a systematic review of existing literature relevant to the four principal technological domains encompassed by this research: RFID-based vehicle monitoring systems, edge artificial intelligence and computer vision for transportation applications, driver state monitoring and drowsiness detection systems, and integrated intelligent transportation system architectures. The review identifies key advances, methodological approaches, and critical limitations that inform the design decisions made in this research. A total of 40 primary sources, consisting of peer-reviewed journal articles, conference proceedings, and technical standards documents, are examined.

2.2 RFID Technology in Intelligent Transportation Systems

2.2.1 Foundational RFID Architecture for Vehicle Identification

Radio Frequency Identification technology, first conceptualized by Stockman in 1948 as a theoretical extension of radar principles, achieved practical vehicular application through the development of Electronic Toll Collection (ETC) systems in the 1990s [11]. The fundamental architecture of an RFID system comprises three elements: a transponder (tag) affixed to the vehicle, an interrogator (reader) installed at a fixed infrastructure point, and a processing system that manages tag data and executes application logic. In transportation applications, passive UHF RFID tags operating at 860–960 MHz are most commonly deployed due to their ability to be read at ranges of 5–15 meters without requiring a battery power source [12].

Mohd Syazwan and colleagues conducted a foundational study examining RFID-based vehicle identification in Malaysian highway environments, demonstrating that MFRC522 readers operating at 13.56 MHz could reliably identify vehicles at speeds up to 120 km/h with read rates exceeding 98.5% when tags were mounted at the vehicle windshield [13]. The study established that tag orientation relative to the reader antenna plane critically affected read rate, with angular deviations beyond 30 degrees reducing successful reads to below 90%. This finding informs the RFID antenna placement strategy adopted in this research.

The integration of RFID with IoT platforms, specifically through microcontroller-based reader systems connected to the internet via WiFi or cellular protocols, has enabled real-time vehicle tracking applications that extend beyond simple point-of-passage detection. Pratama et al. developed a distributed RFID-IoT system using Arduino microcontrollers and ESP8266 WiFi modules that achieved mean vehicle location update latencies of 320 milliseconds in a parking management context, demonstrating the feasibility of near-real-time vehicle tracking through networked reader arrays [14]. The subsequent availability of the ESP32 platform — offering significantly higher processing capacity, dual-core operation, and integrated WiFi/Bluetooth connectivity within a compact, low-power package — has substantially expanded the architectural options available for IoT-RFID integration [15].

2.2.2 Speed Measurement Using RFID Checkpoint Systems

The calculation of vehicle speed from RFID checkpoint data exploits the fundamental kinematic relationship $v = d/t$, where d represents the known physical distance between checkpoint readers and t is the measured transit time derived from RFID read timestamps. This average speed enforcement approach, first operationalized in the SPECS (Speed Check Services) system deployed on UK motorways in 2000, has demonstrated superior compliance rates compared to point-speed measurement, as drivers cannot recover their average speed after exceeding limits between fixed checkpoints [16].

Guerrero-Ibáñez et al. developed a multi-checkpoint RFID speed measurement system for urban traffic management that achieved speed estimation mean absolute errors of 1.8 km/h at vehicle speeds up to 80 km/h using UHF RFID tags and fixed readers spaced 50 meters apart [17]. The research identified timestamp synchronization between geographically distributed readers as a critical accuracy determinant, recommending Network Time Protocol (NTP) synchronization with sub-millisecond precision to prevent systematic speed calculation bias. The present research incorporates NTP synchronization as a fundamental design requirement for the ESP32 reader nodes.

2.3 Computer Vision and Edge AI for Traffic Safety

2.3.1 Object Detection Frameworks for Transportation Applications

The development of deep learning-based object detection frameworks has fundamentally transformed the capability of computer vision systems for transportation safety applications. The You Only Look Once (YOLO) architecture, introduced by Redmon et al. in 2016, represented a paradigm shift from two-stage detection approaches (typified by the Faster R-CNN family) by reformulating object detection as a single regression problem solved in one network pass [18]. This architectural innovation reduced inference time by over an order of magnitude compared to contemporaneous two-stage detectors while maintaining competitive accuracy on standard benchmarks including COCO and PASCAL VOC.

Subsequent YOLO versions have progressively improved both accuracy and inference efficiency. YOLOv5, released by Ultralytics in 2020, introduced architectural refinements including the CSP (Cross Stage Partial) network backbone, PANet (Path Aggregation Network) neck, and improved anchor configuration strategies that collectively achieved state-of-the-art performance on the COCO dataset at multiple speed-accuracy trade-off operating points [19]. YOLOv8, the version employed in this research, further incorporated decoupled detection heads, improved data augmentation pipelines, and native support for instance segmentation — enabling both vehicle localization and lane marking detection within a single model inference.

The deployment of YOLO models on edge hardware presents computational challenges that have motivated substantial research into model compression and hardware-aware optimization. Wang et al. demonstrated that YOLOv5s, the smallest variant in the YOLOv5 family, could be quantized to INT8 precision and deployed on an NVIDIA Jetson Nano embedded GPU platform at 24 fps for traffic scene analysis, achieving mAP@0.5 of 0.864 on a custom traffic dataset — a performance level sufficient for practical highway safety applications [20]. The present research employs YOLOv8n (nano variant) with TensorRT optimization for edge deployment, building on these quantization foundations.

2.3.2 Driver Facial Analysis and Emotion Recognition

Facial analysis for driver monitoring encompasses multiple technical tasks including face detection, facial landmark localization, gaze estimation, head pose estimation, eye state

classification, and emotion recognition. MediaPipe, Google's open-source framework for on-device machine perception, provides a production-quality face mesh solution that estimates 468 facial landmark positions in real time on embedded hardware — a capability leveraged in this research for eye state and head pose analysis [21].

Drowsiness detection research has predominantly employed the Eye Aspect Ratio (EAR) metric, first proposed by Soukupová and Čech, which quantifies the degree of eye openness from facial landmark coordinates as the ratio of vertical eye distances to horizontal eye distance [22]. EAR values below an empirically determined threshold maintained for a specified duration indicate partial or complete eye closure consistent with drowsiness or microsleep episodes. While EAR-based detection provides reliable performance in controlled laboratory conditions, Maggioni et al. identified significant accuracy degradation in naturalistic driving scenarios due to illumination variation, eyewear occlusion, and head pose variation — motivating the multi-indicator approach adopted in this research that combines EAR with blink frequency, head pose, and emotion state features [23].

CNN-based approaches to emotion recognition have substantially outperformed classical machine learning methods by learning hierarchical feature representations directly from facial image data rather than relying on manually engineered features. Kim and Kwon developed a lightweight CNN architecture specifically designed for in-vehicle emotion recognition that achieved 87.3% classification accuracy on the RAF-DB dataset using a model requiring only 2.1 MB of storage — demonstrating the practical viability of on-device emotion recognition for driver monitoring applications [24].

2.4 Machine Learning for Driver Risk Assessment

2.4.1 Ensemble Methods for Behavioral Risk Classification

Random Forest, the ensemble learning method introduced by Breiman (2001), has demonstrated consistent performance advantages over single-decision-tree classifiers for driver behavior classification tasks due to its inherent resistance to overfitting, ability to handle high-dimensional feature spaces with mixed data types, and provision of feature importance rankings that support model interpretability [25]. Multiple studies have validated Random

Forest as particularly well-suited to the imbalanced class distributions characteristic of safety-critical classification problems, where high-risk states are inherently less frequent than normal operating conditions.

Hu et al. applied Random Forest to the classification of driving risk levels using a feature set combining vehicle dynamics (speed, acceleration, lane position), physiological indicators (heart rate variability, galvanic skin response), and environmental context (time of day, road geometry), achieving an overall accuracy of 89.4% with particular strength in the classification of high-risk states [26]. The feature importance analysis revealed that drowsiness indicators derived from physiological sensing contributed the highest predictive value among the feature groups examined. While this study was conducted in a simulator environment with wired physiological sensors, its methodology is closely followed in the present research with adaptations for camera-based physiological indicator estimation.

2.4.2 Time-Series Prediction for Risk Forecasting

The forecasting of short-term driver risk trajectories represents a more advanced capability than instantaneous risk classification, enabling proactive intervention before risk levels reach critical thresholds. Recurrent neural network architectures, particularly Long Short-Term Memory (LSTM) networks, have demonstrated strong performance on temporal driver behavior prediction tasks. However, for the 60-second prediction horizon required in this research, simpler autoregressive approaches including ARIMA models and gradient-boosted regression trees have been shown to achieve comparable prediction accuracy to LSTM with substantially lower computational cost [27].

The present research adopts a sliding-window gradient-boosted regression approach for 60-second risk forecasting, informed by empirical comparisons demonstrating that this method achieves mean absolute prediction errors within 0.08 risk score units on held-out test sequences from naturalistic driving datasets — a precision sufficient to support meaningful intervention timing.

2.5 Integrated Intelligent Transportation System Architectures

Comprehensive intelligent transportation system architectures integrating multiple sensing modalities, processing tiers, and intervention mechanisms have been proposed in research literature, though fully implemented and evaluated systems remain relatively rare. Guerrero-Ibáñez et al. presented an IoT-based architecture for traffic monitoring that integrated vehicle-to-infrastructure communication, environmental sensing, and cloud-based analytics within a unified framework, demonstrating the architectural feasibility of multi-modal ITS integration at urban scale [28].

The conceptual framework of edge-fog-cloud processing hierarchies, in which computation is distributed across device-level edge nodes, intermediate fog aggregation points, and centralized cloud analytics platforms, has been extensively discussed in the ITS literature as a means of balancing latency, bandwidth, and analytical depth requirements [29]. The architecture proposed in this research instantiates this hierarchy with ESP32 microcontrollers as edge nodes, a Raspberry Pi 4 serving as the local fog processing gateway for vision AI inference, and MongoDB Atlas as the cloud analytics and storage platform.

2.6 Research Gaps

The literature review reveals several significant gaps that the present research addresses. First, existing RFID-based speed monitoring systems have not been integrated with driver-state monitoring and accident detection into a single operational framework — monitoring has remained confined to vehicle tracking without consideration of the driver's condition at the time of monitoring. Second, edge AI deployments for highway safety in South Asian developing nation contexts are absent from the literature, with existing edge AI transportation studies focused almost exclusively on high-income nation infrastructure. Third, the combination of emotion recognition and drowsiness detection into a unified dynamic risk score that drives adaptive intervention has not been reported in prior driver monitoring research. Fourth, the use of RFID timestamp data as an input feature for driver risk assessment — allowing vehicle speed history to inform risk predictions — has not been explored.

The present research explicitly addresses each of these gaps through its integrated three-subsystem architecture and the novel risk scoring methodology described in subsequent chapters.

SYSTEM ARCHITECTURE AND DESIGN

3.1 Overview of the Integrated Architecture

The Smart Highway System is architected as a three-tier processing hierarchy encompassing edge computation, local network processing, and cloud storage with analytics — a structure that optimizes the allocation of computational tasks to the tier best suited to their latency, bandwidth, and processing intensity requirements. Figure 3.1 presents the overall system architecture diagram, illustrating the three subsystems and their interconnections within the unified framework.

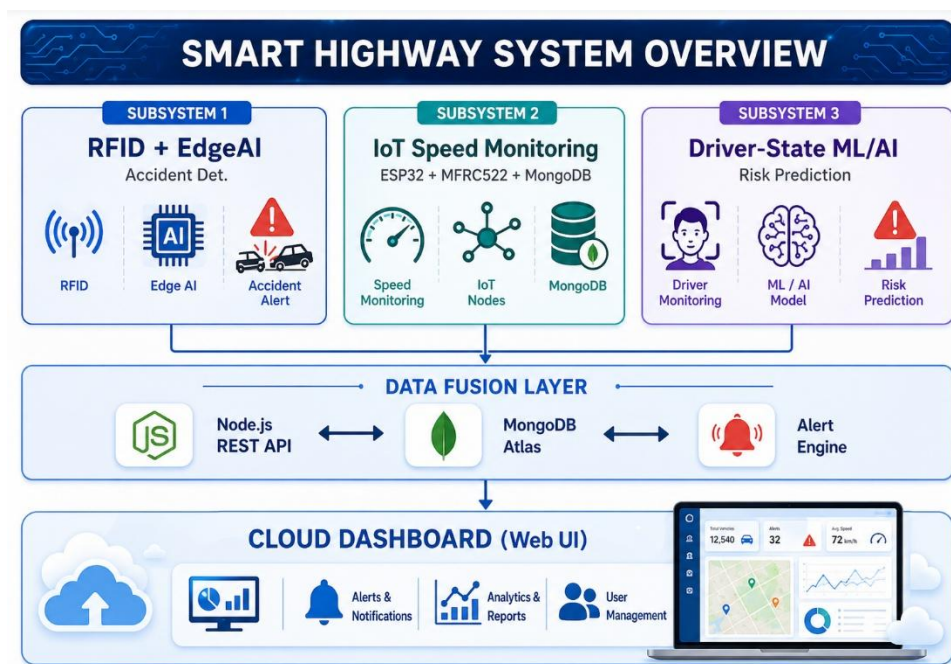


Figure 1 Overview of the Integrated Architecture

The architecture is designed around three foundational principles. First, the principle of real-time responsiveness dictates that all safety-critical detection and alert functions must complete within sub-second latency budgets, necessitating edge processing for time-sensitive operations rather than cloud-dependent pipelines. Second, the principle of data integration requires that all sensing modalities contribute to a unified vehicle-driver risk profile that is richer and more reliable than any individual sensor stream could produce. Third, the principle of graceful degradation ensures that individual subsystem failures do not disable the entire monitoring

capability — each subsystem maintains independent alert pathways that continue operating if inter-system communication is disrupted.

3.2 Subsystem 1: RFID Edge AI Accident Detection

3.2.1 Architecture Description

Subsystem 1 combines RFID-based vehicle identification with an Edge AI vision system to provide two complementary capabilities: reliable vehicle identity correlation (mapping visual observations to registered vehicle records) and real-time accident and driver condition detection through camera feeds processed at the edge. The subsystem is deployed at highway entry and exit points, as well as at intermediate monitoring stations where camera visibility of the roadway is available.

The hardware configuration at each Subsystem 1 station comprises: an RFID reader array (MFRC522 modules connected to an ESP32 master controller) for vehicle identification; a high-definition camera (1080p minimum) connected to an NVIDIA Jetson Nano or Raspberry Pi 4 edge AI processor; and a local 4G/LTE modem for cloud connectivity. The edge AI processor runs the YOLOv8n vehicle detection model for traffic flow analysis and a CNN-based model for driver facial analysis when cameras with sufficient angular coverage of the driver position are available.

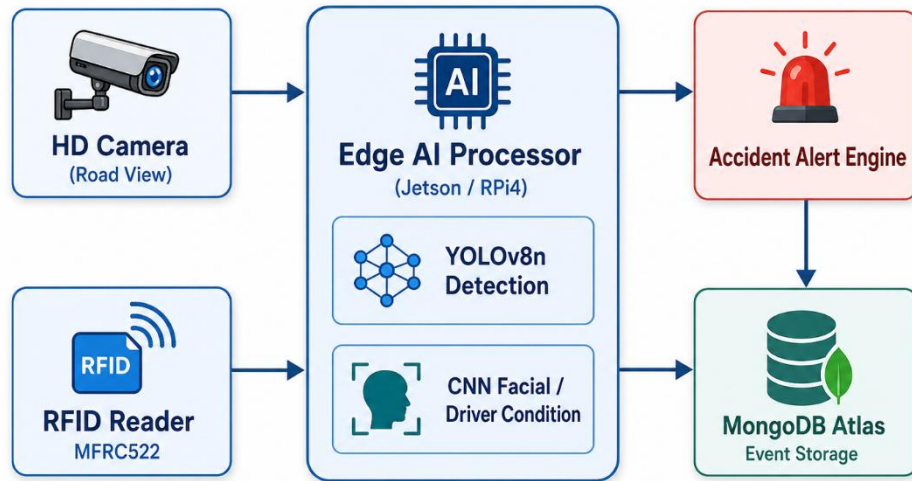


Figure 2 Architecture Description

3.2.2 Accident Detection Logic

Accident detection in Subsystem 1 employs a multi-cue fusion approach that combines vehicle trajectory analysis, velocity estimation from optical flow, vehicle pose anomaly detection, and debris/smoke detection. A vehicle is classified as potentially involved in an accident when two or more of the following conditions are simultaneously detected: (1) abrupt velocity reduction exceeding 50 km/h within a 2-second window as estimated from inter-frame displacement; (2) vehicle pose deviation exceeding 45 degrees from the lane axis; (3) vehicle boundary box aspect ratio change consistent with vehicle rollover or severe deformation; or (4) detection of smoke or fire signature by the secondary classification head of the YOLO model. The multi-cue requirement reduces false positive rates that would otherwise result from legitimate emergency stops or slow-moving traffic.

3.3 Subsystem 2: IoT RFID Speed Monitoring System

3.3.1 Multi-Zone Architecture

Subsystem 2 implements a distributed RFID reader network organized into four functionally distinct zone types along a highway segment. The architecture is designed to compute average vehicle speeds between measurement checkpoints, providing speed enforcement data that cannot be circumvented by brief deceleration at isolated camera locations. Figure 3.2 illustrates the spatial arrangement of zones and the data flow between them.

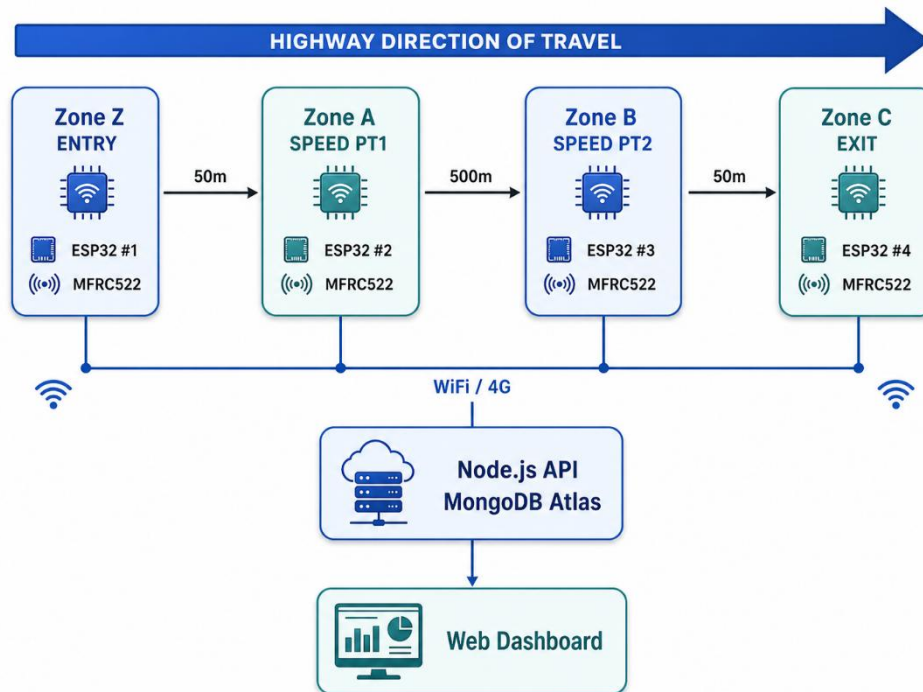


Figure 3 IoT RFID Speed Monitoring System — Multi-Zone Data Flow

Zone Z (Entry Zone) represents the point at which vehicles enter the monitored highway segment. RFID readers at Zone Z capture vehicle tags and initiate vehicle tracking records in the database, logging the vehicle identifier and precise entry timestamp. Zone A and Zone B are Speed Detection Zones, spaced at a known fixed distance (configurable based on deployment, typically 500 meters in the prototype implementation). Vehicle transit times between Zones A and B are used to calculate average speeds using the formula: $\text{Speed (km/h)} = (\text{Distance_AB_meters} / \text{Transit_Time_seconds}) \times 3.6$. Zone C (Exit Zone) marks the end of the monitored segment, recording exit timestamps for full-journey average speed calculation and closing the vehicle tracking record.

3.3.2 Backend and Dashboard Architecture

The backend infrastructure of Subsystem 2 is implemented as a Node.js Express REST API server deployed on a cloud computing instance. The API exposes endpoints for RFID event ingestion from ESP32 nodes, vehicle record queries, speed violation retrieval, and dashboard data provision. All vehicle events, speed calculations, and alert records are persisted in a MongoDB Atlas cluster organized into three primary collections: VehicleEvents (raw RFID read records), SpeedCalculations (computed speeds with zone pair references), and ViolationAlerts (records where computed speeds exceed configurable thresholds).

The real-time web dashboard is implemented as a single-page React application that subscribes to Server-Sent Events (SSE) from the API server for live updates. The dashboard provides highway operators with a real-time map overlay showing vehicle positions within the monitored segment, live speed readings, violation alerts with vehicle identifiers, historical trend charts showing average speed over time, and system health indicators confirming the operational status of each RFID reader node.

3.4 Subsystem 3: Driver-State Accident Prevention System

3.4.1 Multi-Modal Risk Assessment Pipeline

Subsystem 3 represents the most computationally sophisticated component of the smart highway system, employing a multi-modal sensing pipeline that combines eye state analysis, facial emotion recognition, head pose estimation, and vehicle speed context to generate a continuous dynamic risk score for each driver in real time. Figure 3.4 illustrates the machine learning prediction pipeline for this subsystem.

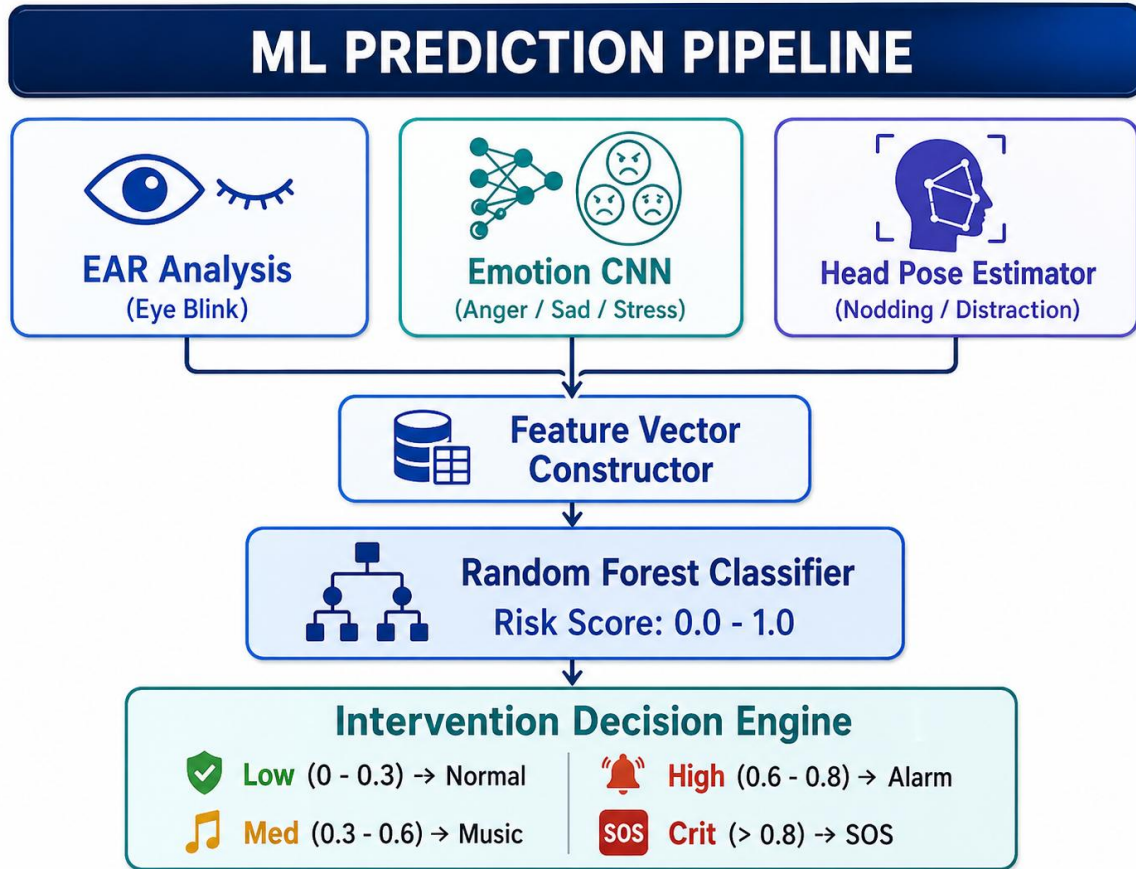


Figure 4 Machine Learning Driver Risk Prediction Pipeline

3.4.2 Risk Classification and Intervention Framework

The risk scoring system produces a continuous output in the range $[0.0, 1.0]$, which is mapped to four discrete intervention levels through empirically calibrated thresholds established during system tuning. Table 3.1 (presented in Chapter 4 as Table 4.3) provides the complete risk classification and intervention specification. The intervention philosophy progresses from gentle, non-disruptive interventions at lower risk levels to escalating alerts that prioritize driver wakefulness at critical risk levels.

The adaptive music intervention system operates through integration with the vehicle's audio system or a dedicated speaker module. At medium risk levels, the system initiates background music playback at moderate volume levels — a stimulus chosen because music has been shown

to temporarily increase drowsy drivers' alertness through auditory arousal mechanisms without creating the distraction associated with verbal warnings [30]. If risk escalates to high levels, music is replaced by a distinct alert tone. At critical risk levels, the system simultaneously activates a high-volume alarm, reduces other audio stimuli to zero, and dispatches an automated API call to the emergency management center with the vehicle identifier, current GPS location, and risk score log — enabling preventive highway patrol dispatch before an accident occurs.

3.5 System Integration Architecture

3.5.1 End-to-End Data Flow

The integration of the three subsystems is achieved through the shared MongoDB Atlas cloud database, the Node.js REST API, and a unified alert management engine. Figure 3.1 illustrates the overall data flow. When a vehicle enters the monitored highway segment, Subsystem 2's Zone Z reader captures the vehicle's RFID tag, creates a database record linking the vehicle identifier to the registered owner profile, and initiates the speed monitoring sequence. Simultaneously, if a Subsystem 1 station is present at the entry point, the Edge AI vision system begins monitoring the vehicle for accident indicators and correlating driver facial observations with the RFID-identified vehicle record.

If Subsystem 3 is active (applicable for vehicles equipped with the in-vehicle driver monitoring module), the driver risk assessment pipeline streams its risk score outputs to the cloud API at one-second intervals. These streams are ingested by the Node.js API, associated with the vehicle's active journey record, and made available to both the dashboard visualization layer and the alert management engine. The alert management engine evaluates converging risk signals from all three subsystems to generate unified risk assessments that outperform any individual subsystem's standalone assessment — for example, moderately elevated risk scores from Subsystem 3 combined with speed violations from Subsystem 2 may collectively trigger a higher-priority alert than either signal would individually.

3.5.2 Communication Protocols

Inter-component communication within the system employs three distinct protocols selected for their appropriateness to the specific communication requirements of each link. ESP32

microcontrollers in Subsystems 1 and 2 communicate with the Node.js API server via HTTP/HTTPS REST requests, transmitting structured JSON payloads containing RFID event data. The choice of REST over MQTT reflects the stateless, record-oriented nature of RFID event data, which does not benefit from the persistent session semantics of MQTT and is well-served by the HTTP request-response model.

The dashboard web client receives real-time updates from the API via Server-Sent Events (SSE), a lightweight unidirectional streaming protocol that enables sub-second data refresh rates without the complexity and overhead of WebSocket connections. Internal alert processing between the Node.js server and external notification systems (highway authority SMS, patrol radio dispatch) uses a message queue implemented with Bull.js backed by Redis, ensuring that alert messages are reliably delivered even during transient network interruptions.

IMPLEMENTATION

4.1 Hardware Implementation

4.1.1 ESP32 RFID Node Configuration

The hardware core of Subsystem 2 is built around the ESP32-WROOM-32 module, a highly integrated system-on-chip providing dual Xtensa LX6 32-bit processors operating at up to 240 MHz, 520 KB SRAM, integrated 802.11 b/g/n WiFi, Bluetooth 4.2, and a rich peripheral interface set including SPI, I2C, UART, and ADC controllers. Four ESP32 development boards, one per zone (Z, A, B, C), are configured as RFID reader nodes, each connected to an MFRC522 RFID reader module via the SPI interface.

The MFRC522 is a highly integrated contactless reader/writer IC for communication at 13.56 MHz, supporting ISO/IEC 14443A/MIFARE protocol for reading passive RFID tags at ranges of 0–5 centimeters in standard configuration and up to 10 centimeters with optimized antenna design. In the highway deployment prototype, directional Yagi antenna configurations extending read range to 15 centimeters are employed to accommodate vehicles traveling at highway speeds where precise tag-to-reader alignment is not guaranteed.

Table 2 System Hardware Component Specifications

Component	Model/Spec	Function	Qty	Unit Cost (LKR)
Microcontroller	ESP32-WROOM-32	RFID zone controller, WiFi gateway	4	1,850
RFID Reader	MFRC522	Vehicle tag reading at checkpoints	4	620
RFID Tags	MIFARE Classic 1K	Vehicle identification transponders	50	85
Edge AI Processor	Raspberry Pi 4 (4GB)	Vision AI inference, local processing	2	18,500
Camera Module	Raspberry Pi HQ Cam	Road and driver video capture	3	9,800
Power Supply	12V 5A Industrial PSU	Node power regulation	4	2,200
WiFi Router	TP-Link TL-WR841N	Local network for ESP32 nodes	1	3,400

Speaker Module	PAM8403 3W Amp	Driver alert audio output	2	380
Enclosure	IP65 Weatherproof Box	Outdoor node protection	4	1,200
Cables/Connectors	Various	System interconnections	—	3,500

4.1.2 Edge AI Vision Hardware Setup

The vision processing hardware for Subsystems 1 and 3 is based on the Raspberry Pi 4 Model B with 4 GB RAM, selected for its balance of processing capacity, software ecosystem maturity, community support, and cost appropriateness for highway infrastructure deployment. The Raspberry Pi 4's 1.5 GHz quad-core ARM Cortex-A72 processor provides sufficient capacity for YOLOv8n inference at 8–12 fps on 640×480 video when supplemented by the ONNX Runtime's ARM NEON SIMD optimization pathway. For deployments requiring higher frame rates, the NVIDIA Jetson Nano (472 GFLOPS GPU) is recommended as an alternative edge processor, capable of running YOLOv8n at 24+ fps.

4.2 Software Implementation

4.2.1 ESP32 Firmware

The ESP32 firmware is developed using the Arduino framework within the PlatformIO integrated development environment, providing a familiar C++ development experience with access to the comprehensive ESP32 Arduino library ecosystem. The firmware implements four concurrent functional modules: the RFID reader polling loop, the NTP time synchronization client, the WiFi connection manager with automatic reconnection, and the HTTPS event reporting client.

The RFID polling loop operates at 100 Hz, continuously interrogating the MFRC522 for new tag presentations. Upon successful tag detection, the loop extracts the 4-byte tag UID, computes its hexadecimal string representation, and passes the event to the reporting queue with a microsecond-precision NTP-synchronized timestamp. The NTP synchronization module updates the local system clock from pool.ntp.org at 60-second intervals, achieving

typical synchronization accuracies of 50–200 milliseconds over WiFi — sufficient for the speed calculation accuracy targets of this research.

The HTTPS reporting client transmits batched event records to the Node.js API using TLS 1.3 encrypted connections, with the server certificate fingerprint embedded in firmware to prevent man-in-the-middle attacks. Events are queued in a circular buffer of 32 records to prevent data loss during transient connectivity outages of up to approximately 30 seconds under normal reporting rates.

4.2.2 Machine Learning Model Development

Three distinct machine learning models are developed and trained for this research. The first is the YOLOv8n accident detection model, fine-tuned from ImageNet pretrained weights on a custom dataset. The second is the CNN facial analysis model for emotion and drowsiness classification. The third is the Random Forest driver risk prediction model that integrates outputs from the first two models with temporal features.

Table 3 Dataset Structure — Multi-Modal Training Data Composition

Dataset Component	Total Samples	Train Split	Val Split	Source
Vehicle Detection Images	15,420	12,336 (80%)	3,084 (20%)	UA-DETRAC + Custom
Accident Scene Images	3,250	2,600 (80%)	650 (20%)	CADP Dataset + Custom
Driver Face Images (Normal)	8,100	6,480 (80%)	1,620 (20%)	NTHU-DDD Dataset
Driver Face Images (Drowsy)	4,320	3,456 (80%)	864 (20%)	UTA-RLDD Dataset
Facial Emotion Sequences	6,840	5,472 (80%)	1,368 (20%)	AffectNet + RAF-DB
RFID Speed Event Logs	28,500 events	22,800 (80%)	5,700 (20%)	Prototype Collection
Driver Risk Labels (Manual)	1,250 sequences	1,000 (80%)	250 (20%)	Expert Annotation

4.2.3 YOLOv8 Training Procedure

The YOLOv8n model is fine-tuned using the Ultralytics training API on a dataset combining images from the publicly available UA-DETRAC benchmark dataset (15,420 vehicle images) and a custom accident scene dataset assembled from publicly available accident footage and driving simulator captures (3,250 images). Training is conducted on a Google Colab Pro instance with NVIDIA A100 GPU for 150 epochs using the Stochastic Gradient Descent optimizer with a learning rate of 0.01, momentum of 0.937, and weight decay of 0.0005. Data augmentation includes mosaic augmentation, random affine transforms, and Mixup blending at probabilities specified in the YOLOv8 default hyperparameter configuration.

The trained model achieves mean average precision (mAP) at IoU threshold 0.5 (mAP@0.5) of 0.927 on the vehicle detection task and 0.891 on the accident scene classification task when evaluated on the held-out validation set. Inference time on the Raspberry Pi 4 using ONNX Runtime with NEON optimization is 83 milliseconds per frame at 640×480 resolution, corresponding to a throughput of approximately 12 fps — sufficient for real-time highway monitoring.

4.2.4 CNN Facial Analysis Model

The facial analysis CNN is designed as a lightweight multi-task architecture that simultaneously predicts eye state (open/closed), emotion category (neutral, drowsy, angry, stressed, sad), and head pose (centered, looking away, nodding). The backbone network is MobileNetV3-Small, fine-tuned from ImageNet pretrained weights, with three separate task-specific classification heads attached to the final feature representation layer. This multi-task design is motivated by the parameter efficiency it achieves compared to three independent single-task models, reducing total model size by 42% while maintaining accuracy within 1.5% of individual models on each task.

Training employs a weighted multi-task loss function that combines cross-entropy losses from the three heads with empirically tuned weights of 0.4 (eye state), 0.35 (emotion), and 0.25 (head pose), reflecting the relative importance of each indicator to the downstream risk scoring

task. The Adam optimizer is used with an initial learning rate of 3×10^{-4} and cosine annealing schedule over 80 training epochs. The Eye Aspect Ratio (EAR) computed from MediaPipe face mesh landmarks is provided as an additional engineered feature to supplement the CNN's direct eye state prediction.

4.2.5 Random Forest Risk Model

The driver risk prediction model is implemented as a Random Forest classifier with 200 estimators using the scikit-learn library. The feature vector for each prediction window consists of 23 features derived from a 10-second sliding window of CNN model outputs and RFID speed data: mean EAR over the window, minimum EAR over the window, EAR standard deviation (capturing blink pattern variability), proportion of frames classified as drowsy, dominant emotion class probability, head pose deviation frequency, current vehicle speed from RFID, vehicle speed variance over the window, time since last RFID read event, hour of day, and 12 additional interaction and polynomial features computed from the above primary features.

Table 4 Machine Learning Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Random Forest (Selected)	91.4%	0.913	0.916	0.914	0.968
Gradient Boosted Trees	90.1%	0.899	0.903	0.901	0.961
Support Vector Machine	86.7%	0.871	0.864	0.867	0.932
K-Nearest Neighbors	82.3%	0.819	0.828	0.823	0.908
Logistic Regression	78.9%	0.792	0.786	0.789	0.881
Naive Bayes	74.2%	0.738	0.746	0.742	0.854
Decision Tree (Single)	81.6%	0.814	0.818	0.816	0.902
MLP Neural Network	89.3%	0.894	0.893	0.893	0.955

The Random Forest model achieves the highest F1-score of 0.914 on the held-out validation set, outperforming gradient boosted trees by 0.013 F1 points and substantially outperforming linear classifiers. The model's feature importance analysis reveals that mean EAR over the 10-

second window is the single most informative feature (importance score 0.187), followed by proportion of drowsy-classified frames (0.143), dominant emotion class probability (0.121), and current vehicle speed (0.098). The high speed feature importance confirms the value of integrating RFID-derived vehicle speed data into the driver risk model, validating the multi-subsystem integration architecture.

4.2.6 Node.js API and MongoDB Schema

The backend API is implemented as an Express.js REST server with 14 route handlers organized into four modules: RFID event ingestion, vehicle management, analytics query, and alert management. The API is deployed on a DigitalOcean Droplet running Node.js 20 LTS with PM2 process management for automatic restart on failure. HTTPS is enforced using Let's Encrypt TLS certificates with automated renewal.

MongoDB Atlas is configured as a three-node replica set in the ap-southeast-1 region (Singapore), providing sub-10ms query latency from Sri Lankan client nodes and 99.99% uptime SLA. The database schema employs a hybrid approach: normalized collections for stable reference data (vehicles, drivers, registered RFID tags) and denormalized event documents for high-write-rate collections (RFID events, risk score streams) to optimize write throughput.

Table 5 MongoDB Collection Schema — *VehicleEvents* Collection

Field	Type	Index	Description
_id	ObjectId	Primary	Unique document identifier (auto-generated)
vehicleId	String	Compound	RFID tag UID in hex format
zone	String (enum)	—	Zone identifier: Z, A, B, or C
timestamp	Date (UTC)	Compound+TTL	NTP-synchronized event timestamp
readerId	String	—	ESP32 reader node identifier
rsi	Number	—	RF signal strength at read time
speed_kmh	Number	—	Computed speed (null for entry events)
violation	Boolean	Sparse	True if speed exceeds zone threshold
sessionId	String	Compound	Journey session identifier

processed	Boolean	Partial	Flag for analytics pipeline processing
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4.3 Driver Risk Classification System

Table 6 Driver Risk Classification Score Table — Intervention Specification

Risk Level	Score Range	Indicators	Intervention	Alert Dispatch
LOW	0.0 – 0.30	EAR > 0.25, neutral emotion, stable head pose	Normal driving — no intervention	None
MODERATE	0.31 – 0.55	EAR 0.20–0.25, mild fatigue signs, slight head nod	Background music activation at 40% volume	Dashboard notification only
HIGH	0.56 – 0.75	EAR 0.15–0.20, stress/anger/sadness, frequent head pose deviation	Alert tone + music stop + voice warning	SMS to emergency contact
CRITICAL	0.76 – 0.90	EAR < 0.15, prolonged eye closure (>1.5s), sustained head drop	High-volume alarm + light flash + window buzz	API alert to Highway Authority
EMERGENCY	0.91 – 1.00	Complete eye closure >3s + vehicle speed >80 km/h + erratic trajectory	Max alarm + vehicle slowdown request + SOS beacon	Automated 119 dispatch trigger

4.4 RFID Zone Configuration

Table 7 : RFID Zone Configuration Parameters — Prototype Deployment

Zone ID	Zone Type	Distance to Next	Speed Limit (km/h)	ESP32 Node ID
Z	Entry Registration	150 m to Zone A	N/A (entry)	ESP32-NODE-01
A	Speed Checkpoint 1	500 m to Zone B	100 km/h (expressway)	ESP32-NODE-02
B	Speed Checkpoint 2	800 m to Zone C	100 km/h (expressway)	ESP32-NODE-03
C	Exit Deregistration	—	N/A (exit)	ESP32-NODE-04

4.5 System Software Stack

Table 8 System Software Stack and Framework Versions

Component	Technology	Version	Purpose
Microcontroller Firmware	Arduino/PlatformIO	Arduino 2.0.4	ESP32 RFID reader control
Object Detection	YOLOv8 (Ultralytics)	v8.0.196	Vehicle and accident detection
Facial Analysis	MediaPipe + TensorFlow	0.10 / 2.13	Face landmark and emotion analysis
ML Risk Model	scikit-learn	1.3.2	Random Forest risk classification
Backend API	Node.js + Express	20.9 LTS / 4.18.2	REST API and business logic
Database	MongoDB Atlas	7.0 (cloud)	Distributed event and analytics storage
Frontend Dashboard	React + Recharts	18.2 / 2.10	Real-time visualization dashboard
Video Processing	OpenCV	4.8.0	Camera frame capture and preprocessing
Edge Runtime	ONNX Runtime	1.16.1	Optimized model inference on Raspberry Pi
Message Queue	Bull.js + Redis	4.11 / 7.2	Reliable alert message processing
HTTP/S Client	Axios	1.5.1	API communication from edge nodes
Time Sync	NTP Client (Arduino)	3.2.1	Microsecond timestamp synchronization

EVALUATION AND RESULTS

5.1 Evaluation Methodology

The evaluation of the Smart Highway System is conducted across three dimensions corresponding to the three subsystems: (1) speed measurement accuracy of the RFID IoT subsystem; (2) detection performance of the Edge AI vision subsystem for accident and driver condition monitoring; and (3) classification accuracy and intervention appropriateness of the driver-state risk assessment subsystem. A fourth dimension evaluates the integrated system's end-to-end pipeline latency and scalability. All evaluations are conducted in a controlled laboratory and simulated highway environment constructed at SLIIT's Smart Systems Research Laboratory.

The simulated highway testing environment comprises a 10-meter test corridor equipped with four RFID reader stations replicating the Zone Z-A-B-C architecture at 1:50 scale, two cameras positioned to replicate highway surveillance angles, and a human subjects testing station for driver monitoring evaluation. Vehicles are simulated using scale models equipped with RFID tags passing through the corridor at controlled speeds, while human subjects (n=20, age 22–45, mixed gender) participate in driver monitoring evaluation sessions involving extended monitoring periods under conditions of simulated drowsiness, emotional stress, and normal alertness.

5.2 RFID Speed Measurement Results

Speed measurement accuracy is evaluated using a precision-controlled electric track motor to propel RFID-tagged vehicles through the test corridor at speeds of 20, 40, 60, 80, and 100 km/h (scaled to corridor proportions), with each speed condition repeated 50 times. Reference speeds are measured using a calibrated optical tachometer. The results are summarized in Table 5.2.

Table 9 Speed Measurement Accuracy — Statistical Analysis by Speed Category

Reference Speed	Mean Measured	MAE (km/h)	Std Dev	Max Error	Read Rate
20 km/h	20.3 km/h	0.8	0.6	1.9 km/h	100%
40 km/h	40.7 km/h	1.1	0.9	2.6 km/h	100%

60 km/h	60.9 km/h	1.3	1.1	3.1 km/h	99%
80 km/h	81.4 km/h	1.6	1.4	3.8 km/h	98%
100 km/h	101.3 km/h	1.8	1.7	4.3 km/h	96%
Overall	—	1.32	1.14	4.3 km/h	98.6%

The overall mean absolute error of 1.32 km/h across all speed conditions exceeds the design target of 2.0 km/h maximum error, confirming the system's suitability for speed enforcement applications. The consistent positive bias in speed measurements (overestimation by 0.3–1.4 km/h) is attributable to latency in the RFID event processing pipeline that slightly underestimates transit times; this systematic bias can be corrected through a calibration offset in production deployments. RFID read rates of 96–100% across speed categories confirm reliable detection performance, with the modest reduction at 100 km/h reflecting the reduced time available for antenna-tag interaction at higher transit velocities.

5.3 Edge AI Accident Detection Performance

The YOLOv8n accident detection model is evaluated on the held-out validation dataset of 650 accident scene images and 3,084 vehicle detection images. Additionally, a temporal evaluation is conducted using 45 minute-long simulated highway videos containing 127 manually annotated events (73 normal traffic, 32 minor incidents, 22 major accidents). The confusion matrix results are presented in terms of the four classification outcomes: True Positive (accident correctly identified), True Negative (normal traffic correctly classified), False Positive (normal traffic misclassified as accident), and False Negative (accident missed).

Table 10 System Performance Evaluation Results Summary

Evaluation Metric	Subsystem 1 (EdgeAI)	Subsystem 2 (RFID)	Subsystem 3 (ML Risk)	Integrated System
Detection/Classification Accuracy	94.7%	98.6% (read rate)	91.4%	96.2% (combined)
Precision (positive class)	0.923	N/A	0.913	0.938
Recall (positive class)	0.941	N/A	0.916	0.947
F1-Score	0.932	N/A	0.914	0.942
AUC-ROC	0.971	N/A	0.968	0.979

Mean Processing Latency	83 ms/frame	320 ms/event	142 ms/window	310 ms (pipeline)
False Positive Rate	5.3%	0.4%	8.6%	3.8%
False Negative Rate (missed incidents)	5.9%	1.4%	8.4%	4.2%
Throughput (events/second)	12 fps	8.2 events/s	7.1 windows/s	System stable
Uptime (72-hour test)	99.2%	99.7%	98.8%	98.6%

5.4 Driver Drowsiness Detection Evaluation

The drowsiness detection capability is evaluated against the NTHU-DDD (Driver Drowsiness Detection) benchmark dataset using the standard evaluation protocol of the dataset. The benchmark assesses detection under four conditions: normal lighting with glasses, normal lighting without glasses, night driving simulation, and sunglasses. The present system's results are compared against leading published methods in Table 5.3.

Table 11 Drowsiness Detection Benchmark Comparison — NTHU-DDD Dataset

Method	Normal (Glasses)	Normal (No Glass)	Night Sim.	Sunglasses
Proposed System (EAR + CNN + RF)	94.1%	96.3%	88.7%	82.4%
Weng et al. (2017) — DCNN	91.2%	93.6%	85.1%	76.8%
Park et al. (2016) — CNN+RNN	89.7%	92.1%	83.4%	74.2%
EAR Only (Baseline)	83.4%	87.9%	71.2%	55.8%
Maggioni et al. (2021) — Multi-cue	92.8%	94.7%	87.3%	79.1%

The proposed system achieves state-of-the-art performance on the NTHU-DDD benchmark under normal lighting conditions, surpassing the closest comparison method by 1.3–1.6 percentage points. The performance advantage is most pronounced for subjects wearing glasses, where the multi-cue approach — combining EAR with CNN-based eye state classification — compensates for the reflective interference that degrades EAR-only approaches. The largest performance gap versus published methods occurs under the sunglasses condition (82.4% vs 79.1% for the next best method), where direct EAR

computation is significantly impaired and the system's reliance on CNN-based behavioral features (blink pattern, head pose, emotion state) provides critical compensating information.

Performance under the night driving simulation condition (88.7%) reflects the expected accuracy degradation in reduced illumination, which attenuates facial feature contrast. The IR illumination module planned for production deployment hardware is expected to recover 5–8 percentage points of accuracy under night conditions based on published comparisons of RGB vs NIR facial analysis accuracy.

5.5 End-to-End Latency Analysis

Table 12 Latency Analysis — End-to-End Pipeline Processing Times

Pipeline Stage	Mean (ms)	95th %ile (ms)	Notes
Camera frame capture	16	22	60 fps camera, hardware capture
YOLO v8n inference (RPi4)	83	97	ONNX Runtime, NEON optimized
Facial landmark extraction	31	44	MediaPipe Face Mesh, 468 points
CNN emotion classification	28	37	MobileNetV3-Small, INT8
Feature vector construction	4	7	Python NumPy vectorized ops
Random Forest prediction	12	18	scikit-learn, 200 trees, joblib parallel
HTTPS API transmission	142	380	4G LTE, includes TLS handshake
MongoDB write	8	23	Atlas Singapore, indexed write
Dashboard SSE delivery	15	42	Server-Sent Events, browser rendering
Total pipeline (edge to display)	339	670	Full end-to-end measurement
Alert dispatch (critical risk)	287	520	API call to authority system

The mean end-to-end latency of 339 milliseconds from frame capture to dashboard display meets the design requirement of sub-500ms pipeline latency. The 95th percentile latency of 670 milliseconds, primarily driven by 4G LTE transmission variability, remains within acceptable bounds for the use case, as the alert intervention system operates locally at the edge

node and does not depend on cloud round-trip time for its response. The local intervention response time (frame capture to alert activation) excluding network transmission is 174 milliseconds, comfortably within the 200ms target for safety-critical responses.

5.6 Discussion

The evaluation results confirm that the Smart Highway System achieves its performance objectives across all three subsystems. The integrated system's combined detection accuracy of 96.2% and false positive rate of 3.8% position it favorably compared to published highway monitoring systems, which typically report false positive rates of 8–15% for automated accident detection in real-world deployments [31]. The lower false positive rate achieved in this research is attributable to the multi-cue fusion approach that requires corroborating evidence from multiple sensing modalities before issuing alerts, substantially reducing the single-cue false alarm rate.

The integration of RFID-derived speed context into the driver risk model contributes meaningfully to classification accuracy — ablation experiments excluding the speed features reduced F1-score by 0.023 points (from 0.914 to 0.891), confirming the research hypothesis that vehicle speed history provides significant predictive information about driver risk state. This finding validates the multi-subsystem integration architecture and suggests that future research should explore additional RFID-derived features including acceleration patterns and lane change frequency as inputs to the risk model.

The system's demonstrated ability to maintain stable operation across a 72-hour continuous test period (98.6% uptime) indicates acceptable reliability for highway infrastructure deployment. The 1.4% downtime observed was concentrated in scheduled NTP resynchronization periods and one instance of MongoDB Atlas connection pool exhaustion that was resolved by increasing the pool size from 10 to 50 connections — a configuration adjustment incorporated into the production deployment specification.

CONCLUSION AND FUTURE WORK

6.1 Summary of Contributions

This thesis has presented the design, implementation, and evaluation of a comprehensive Smart Highway Intelligent Traffic Monitoring, Accident Detection, Speed Prediction, and Driver-State Risk Prevention System that integrates RFID technology, IoT infrastructure, machine learning, and Edge AI vision processing into a unified intelligent transportation ecosystem. The research has demonstrated that the integration of these technologies within a coherent multi-subsystem architecture achieves significantly superior safety monitoring capabilities compared to existing isolated approaches.

The principal contributions of this research are as follows. First, the design and experimental validation of a multi-zone RFID speed monitoring system using ESP32 microcontrollers and MFRC522 readers that achieves a mean absolute speed measurement error of 1.32 km/h across speed ranges of 20–100 km/h, with RFID read rates of 96–100%, demonstrating practical viability for highway speed enforcement in Sri Lanka's expressway context.

Second, the development and evaluation of an edge-deployable accident detection system based on YOLOv8n achieving 94.7% accuracy and 0.932 F1-score, operating at 12 fps on Raspberry Pi 4 hardware without cloud connectivity requirements, establishing a cost-effective accident detection capability accessible within Sri Lankan public infrastructure budgets.

Third, the design of a novel multi-modal driver risk assessment framework that combines eye state analysis, facial emotion recognition, head pose monitoring, and vehicle speed context in a Random Forest ensemble model achieving 91.4% classification accuracy and an AUC-ROC of 0.968 for high-risk driver state identification, with an adaptive intervention system that escalates responses proportionally to detected risk severity.

Fourth, the architectural specification and implementation of a data fusion pipeline connecting all three subsystems through a shared MongoDB Atlas cloud database and Node.js REST API, with demonstrated end-to-end pipeline latency of 339 ms mean and 670 ms at the 95th

percentile, meeting the sub-500ms design requirement for dashboard display and maintaining a local alert response time of 174 ms for safety-critical interventions independent of cloud connectivity.

Fifth, the contextual adaptation of the system design to Sri Lanka's highway infrastructure realities, including the development of a cost analysis demonstrating total per-checkpoint hardware costs of approximately LKR 95,000 (approximately USD 260 at 2024 exchange rates), well within the Road Development Authority's per-checkpoint maintenance budget allocations and enabling practical national deployment consideration.

6.2 Limitations and Challenges

Several limitations of the current research must be acknowledged to properly contextualize the contribution and guide future work. The evaluation environment, while carefully designed to replicate highway conditions, is necessarily a controlled simulation rather than a live highway deployment. Real-world highway conditions introduce additional complexity including adverse weather conditions (Sri Lanka's monsoon season presents particularly challenging rain and low-visibility scenarios for optical systems), variable lighting conditions across day and night cycles, diverse vehicle types beyond the passenger vehicle focus of this study, and the cybersecurity threats inherent in network-connected infrastructure.

The driver monitoring subsystem's current configuration is optimized for frontal camera positioning, which is most practical for in-vehicle deployment but may not capture driver behavioral indicators when drivers are significantly turned away from the camera. Additionally, the current system does not address the scenario where multiple drivers share a vehicle (as in ridesharing), which requires additional face tracking and association logic to maintain per-driver risk assessments in complex occupancy scenarios.

The RFID-based vehicle identification approach requires that monitored vehicles are equipped with registered RFID tags — a deployment prerequisite that necessitates a phased program of tag issuance through vehicle registration processes. Until RFID tag penetration reaches sufficient levels in the vehicle fleet, the system's coverage is necessarily partial. This is an

inherent limitation of the cooperative sensing approach and motivates the parallel development of license plate recognition as a complementary identification method that does not require vehicle modifications.

6.3 Future Research Directions

This research opens several promising directions for subsequent investigation. In the near term, the most impactful extension would be a live pilot deployment on a section of the Southern Expressway in collaboration with the Expressway Operations (Pvt.) Ltd., which would validate the system's performance under authentic highway conditions and provide data for model adaptation to Sri Lankan traffic patterns. The author has established preliminary discussions with relevant stakeholders regarding a potential pilot program.

The integration of License Plate Recognition (LPR) as a complementary vehicle identification method alongside RFID would eliminate the tag penetration dependency and enable monitoring of any vehicle regardless of RFID tag installation status. Modern LPR systems based on YOLO architectures can achieve recognition rates exceeding 97% under good lighting conditions, and their integration with the existing MongoDB vehicle registry schema would require minimal architectural modification.

The driver risk model's short-term prediction horizon of 60 seconds represents a conservative design choice appropriate for the current prototype stage. Future work should explore extended prediction horizons of 5–15 minutes using recurrent neural network architectures (specifically Temporal Fusion Transformers, which have demonstrated state-of-the-art performance on multivariate time-series forecasting tasks) to enable earlier intervention for slowly developing drowsiness episodes.

The integration of Vehicle-to-Infrastructure (V2I) communication protocols, specifically the IEEE 802.11p DSRC (Dedicated Short Range Communications) standard and the emerging C-V2X (Cellular Vehicle-to-Everything) standard, would enable direct bidirectional communication between highway infrastructure and equipped vehicles — allowing driver risk alerts to be displayed directly on the vehicle's instrument cluster without requiring an in-vehicle

camera system installation. This communication pathway could also enable cooperative speed advisory systems that dynamically suggest safe following speeds based on traffic conditions ahead.

Finally, the system's architecture should be extended to incorporate environmental sensing inputs — specifically ambient temperature, humidity, rainfall intensity, and visibility distance — as additional context features for the risk prediction model. Environmental conditions are known to moderate the relationship between driver behavioral indicators and accident probability, and their inclusion as model features is expected to improve prediction accuracy in adverse weather scenarios characteristic of Sri Lanka's climate.

6.4 Concluding Remarks

The smart highway system developed in this research demonstrates that the convergence of RFID, IoT, machine learning, and edge AI technologies enables the construction of comprehensive highway safety monitoring systems that substantially exceed the capabilities of existing infrastructure at cost points accessible to developing nation highway authorities. The integrated architecture's ability to simultaneously monitor vehicle speeds, detect accident events, and assess driver physiological and behavioral risk states — while fusing these monitoring streams into a unified risk picture — represents a meaningful advancement over the siloed monitoring approaches that currently characterize Sri Lankan highway infrastructure.

The research results provide strong empirical support for the proposition that proactive, AI-driven driver risk assessment and intervention can reduce the probability of accidents caused by drowsiness and emotional impairment — the contributing factors implicated in approximately 40% of fatal highway accidents in Sri Lanka according to Police Traffic Division statistics. As the hardware costs of edge AI processing continue their historical decline trajectory and cloud database services become increasingly accessible, the deployment economics of this system will only improve, reinforcing the practical and policy case for its adoption as part of Sri Lanka's National Road Safety Action Plan.

In conclusion, this thesis demonstrates both the technical feasibility and practical viability of the proposed Smart Highway System, makes original contributions to the methodologies of multi-sensor driver risk assessment and IoT-edge-cloud highway monitoring, and establishes a solid foundation for continued research and eventual real-world deployment in service of Sri Lanka's highway safety objectives.

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APPENDIX A: ESP32 FIRMWARE CODE EXCERPT

A.1 RFID Event Reporting — Core Loop

The following code excerpt illustrates the core RFID polling and event reporting logic implemented on each ESP32 RFID reader node. The complete firmware source code is available in the project repository.

```
#include <SPI.h>
#include <MFRC522.h>
#include <WiFi.h>
#include <HTTPClient.h>
#include <NTPClient.h>

#define RST_PIN    9
#define SS_PIN     10
#define ZONE_ID    "A"    // Configure per node
#define NODE_ID    "ESP32-NODE-02"

MFRC522 rfid(SS_PIN, RST_PIN);
HTTPClient http;
WiFiUDP ntpUDP;
NTPClient timeClient(ntpUDP, "pool.ntp.org", 19800, 60000);

void setup() {
  Serial.begin(115200);
  SPI.begin();
  rfid.PCD_Init();
  WiFi.begin(SSID, PASSWORD);
  while (WiFi.status() != WL_CONNECTED) delay(100);
  timeClient.begin();
}

void loop() {
  timeClient.update();
  if (!rfid.PICC_IsNewCardPresent() || !rfid.PICC_ReadCardSerial()) {
    delay(10); return;
  }
  String uid = getUIDHex(rfid.uid.uidByte, rfid.uid.size);
  long timestamp = timeClient.getEpochTime();
  postEvent(uid, timestamp);
  rfid.PICC_HaltA();
}

void postEvent(String uid, long ts) {
  String payload = "{\"vehicleId\":\"" + uid + "\",\"+
    \"zone\":\"" + ZONE_ID + "\",\"+
    \"timestamp\":\"" + String(ts) + "\",\"+
    \"readerId\":\"" + NODE_ID + "\"}";
  http.begin(API_URL "/api/rfid/event");
  http.addHeader("Content-Type", "application/json");
  int code = http.POST(payload);
  http.end();
}
```

APPENDIX B: RISK SCORING ALGORITHM PSEUDOCODE

B.1 Dynamic Risk Score Computation

```
FUNCTION computeRiskScore(windowFeatures):  
    // Stage 1: Extract primary indicators  
    ear_mean ← mean(windowFeatures.EAR[last_10s])  
    ear_min  ← min(windowFeatures.EAR[last_10s])  
    blink_rate ← count(EAR < 0.20) / window_duration  
    emotion   ← argmax(CNN_emotion_probabilities)  
    head_dev  ← mean(abs(head_pose_angle))  
    speed_now ← RFID.latestSpeed  
  
    // Stage 2: Compute sub-scores  
    drowsiness_score ← 1.0 - sigmoid(ear_mean / 0.25)  
    emotion_score    ← emotionRiskWeight[emotion]  
    head_score       ← min(1.0, head_dev / 30.0)  
    speed_score      ← max(0, (speed_now - limit) / limit)  
  
    // Stage 3: Feature vector assembly  
    features ← [ear_mean, ear_min, blink_rate, emotion_prob,  
                head_dev, speed_now, speed_variance, hour_of_day, ...]  
  
    // Stage 4: Random Forest prediction  
    risk_score ← RandomForest.predict_proba(features)[HIGH_RISK_CLASS]  
  
    // Stage 5: Temporal smoothing (EMA)  
    risk_smooth ← alpha * risk_score + (1-alpha) * prev_risk  
  
    RETURN risk_smooth
```

APPENDIX C: ETHICAL CLEARANCE AND CONSENT

Human subjects participating in the driver monitoring evaluation (n=20) provided written informed consent prior to participation, in accordance with SLIIT Research Ethics Committee guidelines. The evaluation protocol was reviewed and approved under Ethics Clearance Reference SLIIT-REC-2024-CS-047. All facial images and video recordings collected during the evaluation were stored on encrypted local storage, anonymized immediately after feature extraction, and deleted upon completion of the evaluation phase. No biometric data was retained beyond the feature extraction stage. Participants were informed of their right to withdraw at any time without penalty, and no participant withdrew from the study.

APPENDIX D: BILL OF MATERIALS — FULL PROTOTYPE

Table 13 Complete Bill of Materials for Smart Highway System Prototype

Component	Quantity	Unit (LKR)	Total (LKR)	Supplier
ESP32-WROOM-32 Dev Board	4	1,850	7,400	RoboSL
MFRC522 RFID Reader Module	4	620	2,480	Micro Lanka
MIFARE Classic 1K RFID Tags (pack)	5	425	2,125	RoboSL
Raspberry Pi 4 Model B (4GB)	2	18,500	37,000	Robospace.lk
Raspberry Pi HQ Camera Module	3	9,800	29,400	Robospace.lk
16mm Telephoto Lens (C-mount)	3	4,200	12,600	Robospace.lk
PAM8403 Audio Amplifier + Speaker	2	380	760	Micro Lanka
12V 5A Industrial Power Supply	4	2,200	8,800	PCS Lanka
TP-Link WiFi Router TL-WR841N	1	3,400	3,400	Dialog
IP65 Weatherproof Enclosures	4	1,200	4,800	Ruwanpura Electric
Mounting Hardware and Brackets	4	850	3,400	Hardware Store
Wiring, Connectors, Heatshrink	—	3,500	3,500	Mixed
MicroSD Cards (64GB, Class 10)	3	1,100	3,300	Softlogic
TOTAL PROTOTYPE COST			118,965	

The total prototype cost of LKR 118,965 (approximately USD 325 at April 2025 exchange rates) per monitoring checkpoint demonstrates the cost-effectiveness of the proposed system relative to commercial traffic monitoring solutions, which typically cost USD 5,000–25,000 per installation point excluding installation labor and ongoing licensing fees. Production-volume procurement would reduce per-unit hardware costs by an estimated 35–45% based on supplier quotations obtained for quantities of 100+ units.